

ANIMALS IN DANGER

Until a century ago, the earth still contained large areas of wilderness, where animals had little or no contact with human beings. Since then, the human population has soared closer to 7.5 billion, and our increasing use of energy and raw materials affects the entire planet. Natural habitats are disappearing, and the earth's biodiversity—the sum total of all living species—is in sharp decline. This is a dangerous situation for humanity, because it reduces the earth's biological resources, and makes the world a less stable place. For animals, the results can be disastrous. Some changes are local ones, which threaten individual species, but others, particularly climate change, are global in their reach.

Habitat change

Humans first started to alter habitats when they discovered fire, but with the start of farming, about 10,000 years ago, habitat destruction rapidly increased. Agriculture has been the main driving force behind deforestation, which swept across the Northern Hemisphere in historical times, and continues in the tropics today. It has also been responsible for the destruction of some of the world's major natural grasslands, and of marshes and other freshwater wetlands that supply many wild animals with their food. In recent times, urbanization has become almost as important as a threat—towns and cities take up space, and the roads between them use up even more.

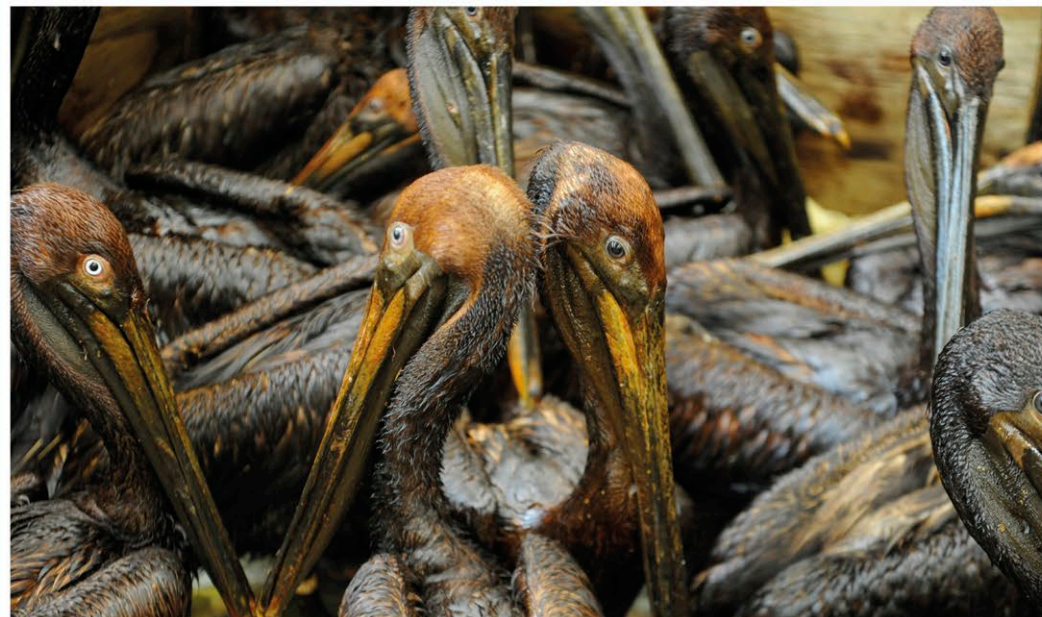
The pattern of habitat change is often as important as its scale. For example, if large areas are preserved, the habitat can often function as before, although on a reduced scale. But if the same amount of habitat is divided up into smaller isolated fragments, the effect on animals is much more severe. This is because many species—

The most problematic chemical pollutants are synthetic organic (carbon-containing) substances, such as solvents, pesticides, and herbicides. Hundreds of thousands of these chemicals now exist, and new kinds are produced every year. Their chemical structure means that they are often absorbed by living tissue, where they are ideally placed to cause the most damage. Some of these substances are toxic to all forms of life, but others are more selective. They are passed on when predators eat their prey, and as a result, they accumulate in species at the top of food chains, such as whales, polar bears, and birds of prey. As well as dissolved pollution, marine animals have to cope with solid plastic waste. Taking decades or even centuries to degrade, this forms huge eddies or gyres in the world's oceans, which can often be hundreds of miles across. Small particles of these plastics are often ingested by animals, weakening them or killing them outright.

Animals are also affected by air pollution, which is created mainly when fuels are burned. Air pollution causes localized problems such as acid rain, which can have a highly damaging effect on freshwater fish. On a much broader scale, it is also responsible for global warming—the biggest environmental change of all.

DEADLY SLICK

Oil spills devastate wildlife. These pelicans were victims of the Deepwater Horizon spill of 2010 in the Gulf of Mexico. Oil is toxic and destroys the waterproofing of plumage; without treatment, affected birds die.



Hunting, fishing, and collecting

Unlike many of the world's other resources, animals can reproduce. This means that—in theory—useful species can be harvested without them ever running out. Unfortunately, many species have been overexploited, with the result that some have died out, while others are now in serious danger.

The list of past casualties from hunting includes the African bluebuck, which died out in about 1800, and the North American passenger pigeon, which became extinct in 1914, despite formerly being one of the most numerous birds in the world. These animals were killed primarily for food, a practice that continues today in a more diverse way, in the bushmeat trade. Once a subsistence activity, the bushmeat trade has recently become a global business, focusing on all kinds of forest animals that can be caught and sold as food. Primates are particularly threatened, but the trade also endangers many other animals, from snakes to pangolins. Animals are also hunted to meet less pressing needs. Elephants are in demand for their ivory, and rhinos for their horns. Tigers are hunted for their fur and body parts, which fetch increasingly inflated prices as the number of surviving animals falls.



AWAITING EXPORT

Packed in tiny wire cages, these parakeets are destined to supply the caged-bird trade. The trade is driven by money from dealers overseas. However, in the birds' native countries, collecting birds may sometimes be the only way of earning a living.



NATURAL PRAIRIE



CEREAL FIELD

FROM PRAIRIE TO FARMLAND

North America's forests and prairies underwent an explosive burst of habitat change with the arrival of European settlers. Both were originally home to a wide variety of animals, such as bears and bison, which were hunted and then displaced as agriculture expanded. Much of the forest has grown back, but the prairies are now almost entirely devoted to cereal farming: the original grassland habitat has largely disappeared.

particularly predators at the top of food chains—need extensive territories to survive. These fragmented habitats are also exposed to more intrusion and disturbance from humans and domestic animals, making it much harder for wild animals to feed and breed.

Pollution

Pollution occurs when chemicals or other agents disrupt natural ecosystems. Sometimes it has a natural origin, but in most cases it is the result of human activity. It can affect animals physically—for example, entangling them in waste, or clogging them with oil—but its chemical effects are often more serious, and harder to identify or predict.